

(1001CJA621621007)

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2021 - 2022)

Test Pattern

JEE(Advanced)

REVIEW TEST

18-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	D	A	C,D	A,B,C	A,D	A,C	A,C	A,B,C,D
	Q.	11	12								
	A.	B,C,D	A,B,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	2.00	84.00	4.00	15.00	0.65 to 0.68	0.65				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	B	A	A,B,C,D	C,D	A,B,C	A,B,C	B,D	A,C
	Q.	11	12								
	A.	A,D	A,B,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	9.00	25.00	4.50	6.00	5.00	5.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	C	C	B,C,D	B,D	A,C,D	A,C,D	A,B,D	A,B,C,D
	Q.	11	12								
	A.	A,B,C,D	A,B,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	5.65 or 5.66	0.11	4.00	1.00	10.00	1.25				

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JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,B,C,D	A,B,C	A,D	A,B,C	B,C,D	A,C,D	A,B,D	A	D
	Q.	11	12								
	A.	C	C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	27.20	18.63	6.00	1.50	158.00	0.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,B,C	B,D	A,B,C	B,D	A,C,D	A,D	A,C,D	D	D
	Q.	11	12								
	A.	D	A								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	77.75	5.00	4.00	5.00	72.00 to 74.00	5.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A,B,C,D	C,D	B,C	A,B	A,C,D	A,B,C	A,C	A	D
	Q.	11	12								
	A.	B	D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	0.52 or 0.53	7.00	1.00	8.00	2.71 or 2.72	2.00				



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PAPER-1

PART-2 : CHEMISTRY

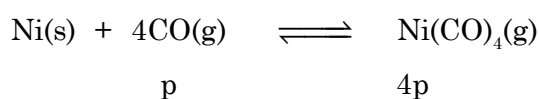
SOLUTION

SECTION-I

1. Ans. (A)

2. Ans.(D)

3. Ans.(B)



$$K_p = \frac{1}{16} = \frac{4p}{p^4} \Rightarrow p = 4 \text{ atm}$$

$\therefore$ Total pressure = $5p = 20 \text{ atm} = 1520 \text{ cm of Hg}$.

4. Ans. (A)

5. Ans.(A,B,C,D)

6. Ans.(C,D)

Line should have higher energy than the energy of line E. Energy of line A($6 \rightarrow 3$) and energy of line D(infinity $\rightarrow 4$) will be greater than the energy of line E($4 \rightarrow 3$).

7. Ans. (A,B,C)

8. Ans.(A,B,C)

9. Ans.(B,D)

$t_{1/2} = 15 \text{ min}$. Volume of H_2O_2 consumed at $t = 60 \text{ min}$ will be $40 + 20 + 10 + 5 = 75 \text{ ml}$. Optical rotation was found to be $\theta = 20^\circ$ at $t = 20 + 15 = 35 \text{ min}$

10. Ans (A,C)

11. Ans.(A,D)

12. Ans.(A,B,D)

$$T_2, \text{ irreversible adiabatic} > T_2, \text{ reversible adiabatic}$$

SECTION-II

1. Ans. (9.00)

2. Ans.(25.00)

3. Ans.(4.50)

The volume of shaded space is $\frac{a^3}{6} = \frac{3^3}{6}$
 $= 4.5 \text{ cm}^3$.

4. Ans. (6.00)

5. Ans.(5.00)

6. Ans.(5.00)

$$PV_m = RT$$

or $P = \frac{R}{V_m} [300 + aV_m^3]$, for minimum P,

$$\frac{dP}{dV_m} = 0$$

$$V_m \Rightarrow \left[\frac{300}{2a} \right]^{1/3} = 5 \text{ L / mole}$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

Sol. $\lim_{x \rightarrow -\infty} \sqrt{x^2 - \frac{4x}{3}} + x + 2$

$$\lim_{x \rightarrow -\infty} \left(\frac{x^2 - \frac{4x}{3} - (x+2)^2}{\sqrt{x^2 - \frac{4x}{3}} - (x+2)} \right)$$

$$\lim_{x \rightarrow -\infty} \frac{-\frac{16}{3}x - 4}{|x| \sqrt{1 - \frac{4}{3x}} - (x+2)}$$

$$\lim_{x \rightarrow -\infty} \frac{x \left(-\frac{16}{3} - \frac{4}{x} \right)}{x \left[-\sqrt{1 - \frac{4}{3x}} - \left(1 + \frac{2}{x} \right) \right]}$$

$$\lim_{x \rightarrow -\infty} -\frac{16}{3(-2)} = \frac{8}{3}$$

2. **Ans. (C)**

Sol. $f(x) + f(y) = f(x)f(y) + f(xy)$

partial differentiation wrt x

$$f'(x) = f'(x)f(y) + f'(xy) \cdot (y)$$

$$\text{put } x = 1$$

$$-2 = -2f(y) + f'(y)y$$

$$\text{get } f(y) = 1 - y^2 \Rightarrow f(x) = 1 - x^2$$

3. **Ans. (C)**

Sol. $e^x(x-3) = k$

$$\text{Let } f(x) = e^x(x-3)$$

$$f'(x) = e^x(x-2)$$

$$k \in (-e^2, 0)$$

4. **Ans. (C)**

Given

$$F(x) = C \sin x \int_0^x \cos t dt + 2 \int_0^x t dt + c \cos^2 x - x^2$$

$$\Rightarrow F'(x) = 0 \therefore F(x) \text{ is a constant function}$$

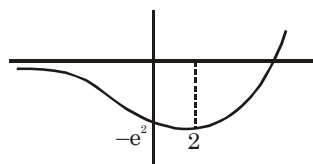
$$\therefore F(x) = F(0) = c$$

Given

$$x^2 - 2x + 3 \geq C \forall x \Rightarrow x^2 - 2x + (3-1) \geq 0 \forall x$$

$$\Rightarrow D \leq 0 \Rightarrow 4 - 4(3-C) \leq 0$$

$$\Rightarrow 1 - 3 + C \leq 0 \Rightarrow C \leq 2$$



$$\text{Now, } \left(\frac{6e^x - e^{2x} - 1}{e^x} \right)^{f(2)} = \left(\underbrace{6 - \left(e^x + \frac{1}{e^x} \right)}_{\leq 4} \right)^c \leq 4^2$$

5. **Ans. (B,C,D)**

$$I = \int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1 + \sqrt[3]{x})} dx \quad \text{Put } x = t^6$$

$$dx = 6t^5 dt$$

$$I = \int \frac{t^6 + t^4 + t}{t^6(1+t^2)} \cdot 6t^5 dt = 6 \int \frac{1+t^3+t^5}{(1+t^2)} dt$$

$$= 6 \left(\int \frac{dt}{1+t^2} + \int t^3 dt \right)$$

$$= \frac{6t^4}{4} + 6 \tan^{-1} t + c = \frac{3x^{2/3}}{2} + 6 \tan^{-1} (x^{1/6}) + c$$

$$\therefore A = \frac{3}{2}, B = \frac{2}{3}, C = 6, D = \frac{1}{6} \Rightarrow AB = 1, AB + C = 7$$

$$\text{Now, } \left(AB + \frac{x}{D} \right)^c = (1+6x)^6$$

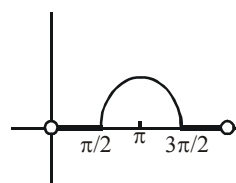
$\therefore$ sum of all coefficients of even power of x

$$= {}^6C_0 + {}^6C_2 \cdot 6^2 + {}^6C_4 \cdot 6^4 + {}^6C_6 \cdot 6^6$$

$$= \frac{(1+6)^6 + (1-6)^6}{2} = \frac{(1+c)^c + (c-1)^c}{2}$$

6. **Ans. (B,D)**

$$\text{Sol. } \cos x [\cos x] = \begin{cases} 0 & 0 < x \leq \frac{\pi}{2} \\ -\cos x & \frac{\pi}{2} < x \leq \pi \\ -\cos x & \pi < x < \frac{3\pi}{2} \\ 0 & \frac{3\pi}{2} \leq x < 2\pi \end{cases}$$



7. Ans. (A,C,D)

Sol. $OM = OC + CM$

$$\Rightarrow x = r\sqrt{2} + r$$

$$\Rightarrow \frac{x}{r} = \sqrt{2} + 1 \Rightarrow \frac{r}{x} = \sqrt{2} - 1$$

$$\Rightarrow \frac{x}{r} = \sqrt{2} + 1 \Rightarrow \frac{r}{x} = \sqrt{2} - 1$$

$$\Rightarrow \frac{dx}{dr} = \sqrt{2} + 1 \text{ and } \frac{dr}{dx} = \sqrt{2} - 1$$

$$\text{and } \frac{d^2x}{dr^2} = 0.$$

Now

$$\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 = (\sqrt{2}+1)^2 + (\sqrt{2}-1)^2 = 2(2+1) = 6$$

$$\& \frac{x+r}{x-r} = \frac{\sqrt{2}+2}{\sqrt{2}} \text{ and } \frac{r-x}{r+x} = \frac{\sqrt{2}-2}{\sqrt{2}}$$

$$\Rightarrow \frac{x+r}{x-r} + \frac{r-x}{r+x} = 2$$

8. Ans. (A,C,D)

Sol. $f(|x|) = \{ |x| - 2, \quad 0 \leq |x| \leq 3$

$$\Rightarrow f(|x|) = \begin{cases} -x-2, & -3 \leq x \leq 0 \\ x-2, & 0 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(|x|)] = \begin{cases} [-x] - 2, & -3 \leq x \leq 0 \\ [x] - 2, & 0 < x \leq 3 \end{cases} \quad \text{Also}$$

$$|f(x)| = \begin{cases} 2, & -3 \leq x \leq 0 \\ 2-x, & 0 < x \leq 2 \\ x-2, & 2 < x \leq 3 \end{cases}$$

$$\Rightarrow [f(x)] = \begin{cases} 2, & -3 \leq x \leq 0 \\ 2+[-x], & 0 < x \leq 2 \\ [x] - 2, & 2 < x \leq 3 \end{cases}$$

$$\text{So } [f(|x|)] + [f(x)] = \begin{cases} [-x], & -3 \leq x \leq 0 \\ [x] + [-x], & 0 < x \leq 2 \\ 2[x] - 4, & 2 < x \leq 3 \end{cases}$$

Now, options can be checked directly without further simplification.

9. Ans. (A,B,D)

Sol. $f(x) = (x-1)\tan^{-1}x - \frac{1}{2}\ln(1+x^2)$

$$f'(x) = \tan^{-1}x - \frac{1}{1+x^2}$$

$$f'(x) > 0 \quad \forall x \geq 1$$

$\Rightarrow f(x)$ is increasing function

$$f''(x) > 0$$

$\Rightarrow f'(x)$ is increasing function $\forall x \geq 1$

$$\lim_{x \rightarrow \infty} f'(x) \rightarrow \frac{\pi}{2} \Rightarrow f'(x) < \frac{\pi}{2} \quad \forall x \geq 1$$

Using LMVT in $[x, x+4], x \geq 1$

$$\frac{f(x+4) - f(x)}{x+4-x} = f'(c) \text{ for } c \in (x, x+4)$$

$$\therefore f(x+4) - f(x) < 2\pi \quad \forall x \in [1, \infty)$$

10. Ans. (A,B,C,D)

Sol. $T = S_1$ and passes through $(1, 0)$

$$y^2 = 2(x-1), \text{ verify all options}$$

11. Ans. (A,B,C,D)

$$\text{Sol. } \sin(\tan^{-1}x) = \frac{x}{\sqrt{1+x^2}},$$

$$\sin(\cos^{-1}x) = \cos(\sin^{-1}x) = \sqrt{1-x^2}$$

Hence, equation reduces to

$$\frac{x}{\sqrt{1+x^2}} = \sqrt{4 - (2\sqrt{1-x^2})^2} = 2|x|$$

$$\frac{x}{\sqrt{1+x^2}} = 2|x| \Rightarrow x=0 \text{ is the only solution}$$

$$\text{A) } \sin^{-1}a + \cos^{-1}a = \frac{\pi}{2} \quad \text{(always true)}$$

$$\text{B) } 2\sin^{-1}a + \cos^{-1}a = \frac{\pi}{2} + \sin^{-1}a = \frac{\pi}{2} \quad \text{(True)}$$

$$\text{C) } \sin^{-1}a + 3\cos^{-1}a = \frac{\pi}{2} + 2\cos^{-1}a = \frac{3\pi}{2} \quad \text{(True)}$$

$$\text{D) } \tan^{-1}a + \cos^{-1}a = \frac{\pi}{2} \quad \text{(True)}$$

12. Ans. (A,B,C)

Sol. $-1 \leq \sum_{k=1}^n 3^k (f(x+ky) - f(x-ky)) \leq 1$

Also, $-1 \leq \sum_{k=1}^{n-1} 3^k (f(x+ky) - f(x-ky)) \leq 1$

$-2 \leq 3^n (f(x+ny) - f(x-ny)) \leq 2$

$-\frac{2}{3^n} \leq (f(x+ny) - f(x-ny)) \leq \frac{2}{3^n}$

$\Rightarrow f(x+ny) - f(x-ny) = 0$

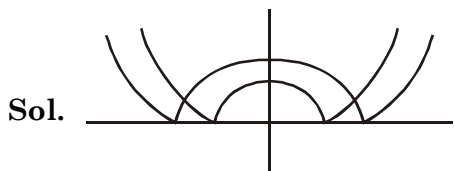
$f'(x+ny) + f'(x-ny) = 0$

$y \rightarrow 0 \Rightarrow f'(x) = 0$

$\Rightarrow f(x)$ is constant function

SECTION-II

1. Ans. (5.65 or 5.66)



Point of intersection when

$x^2 - 1 = 3 - x^2 \Rightarrow x = \sqrt{2}$

$y = x^2 - 1$ & $y = 3 - x^2$

$\left(\frac{dy}{dx}\right)_1 = 2\sqrt{2}$ & $\left(\frac{dy}{dx}\right)_2 = -2\sqrt{2}$

$\tan \theta = \frac{4\sqrt{2}}{1-8} \Rightarrow \theta = \tan^{-1} \left(\frac{4\sqrt{2}}{7} \right)$

2. Ans. 0.11

Sol. $x = 1, f(1) + 2f(1) = 1$

$\Rightarrow f(1) = \frac{1}{3}$

$f(x^2) + 2x^2 f'(x^2) + (x^2 + 1)f'(x) + 2xf(x) = 1$

$x = 1$

$f(1) + 2f'(1) + 2f'(1) + 2f(1) = 1$

$4f'(1) + 1 = 1 \Rightarrow f'(1) = 0$

$2xf'(x^2) + 4x^3 f''(x^2) + 4xf'(x^2) + 2xf'(x)$

$+ (x^2 + 1)f''(x) + 2f(x) + 2f'(x) = 0$

$x = 1$

$2f'(1) + 4f''(1) + 4f'(1) + 2f'(1) + 2f''(1)$

$+ 2f(1) + 2f'(1) = 0$

$6f''(1) + \frac{2}{3} = 0$

$f''(1) = -\frac{1}{9}$

3. Ans. 4.00

$A = \int_0^1 \frac{e^x dx}{\sqrt{1-x^2}}$

Put $x = \sin \theta$

$\Rightarrow A = \int_0^{\pi/2} e^{\sin \theta} d\theta$

Let $I = \int_0^{\pi} (e^{|\sin x|} + e^{|\cos x|}) dx$

↓

Period = $\frac{\pi}{2}$

$\Rightarrow I = 2 \int_0^{\pi/2} (e^{|\sin x|} + e^{|\cos x|}) dx$

$\Rightarrow I = 4 \int_0^{\pi/2} e^{|\sin x|} dx = 4A$

4. Ans. 1.00

Sol. $f'(0^+) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$

$$= \lim_{h \rightarrow 0} \frac{\frac{\sin^{-1}(h^2 + h^3) - h}{h} + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(h^2 + h^3) (h^2 + h^3)}{h^2 (h^2 + h^3)} = 1$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} \frac{\sin^{-1}(h^2 + h^3) - h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{1 - (h^2 + h^3)^2}} \cdot (2h + 3h^2) - 1}{1} = -1$$

5. Ans. (10.00)

Sol. Normal through given point be the axis of parabola

$$y - 2 = -4(x - 1)$$

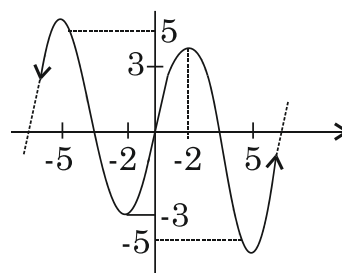
$$4x + y = 6$$

Slope of AB = 4

$$\text{Slope of } \left(\frac{29}{17}, \frac{-14}{17} \right) \text{ and } (1, 2) \text{ is } \frac{-1}{4}$$

6. Ans. 1.25

Sol.



$$f(x) = ax^9 + b \sin x + cx^2 \operatorname{sgn}(x) + \frac{e^x - e^{-x}}{e^x + e^{-x}}; a > 0 \& b, c \in \mathbb{R}$$

is an odd function (check yourself)

$$\Rightarrow f(0) = 0$$

$$f(-5) = 5, f(-2) = -3 \Rightarrow f(5) = -5, f(2) = 3$$

Also, as $x \rightarrow \infty$; $f(x) \rightarrow \infty$ and $x \rightarrow -\infty$;

$$f(x) \rightarrow -\infty$$

Hence no of zeroes of $f(x) = 0$ is 5

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JEE(Advanced)**REVIEW TEST****18-04-2021****JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)****PAPER-2****PART-2 : CHEMISTRY****SOLUTION****SECTION-I**1. **Ans. (A,B,C)**2. **Ans. (A,B,C)**

$$n_{\text{total}} = \frac{PV}{RT} = 0.25 \text{ mole} = 250 \text{ m mole.}$$

$$O_3 = \frac{0.5 \times 200}{2} = 50 \text{ m mole.}$$

$$H_2O_2 = \frac{160 \times 0.5}{2} \text{ m mole} = 40 \text{ m mole}$$

$$\Rightarrow \text{Molarity of } H_2O_2 = \frac{40}{100} = 0.4 \text{ M}$$

$$\text{Volume strength} = 0.4 \times 11.2 = 4.48V$$

3. **Ans. (B,D)**4. **Ans. (A,B,C)**5. **Ans. (B,D)**

$$\text{Heat evolved} = 2.5 \times 250 = 625 \text{ KJ}$$

6. **Ans. (A,C,D)**7. **Ans. (A,D)**8. **Ans. (A,C,D)**9. **Ans. (D)**10. **Ans. (D)**11. **Ans. (D)**12. **Ans. (A)****SECTION-II**1. **Ans. (77.75)**

Benzene and toluene are miscible with each other while mercury dissolves in neither. For benzene and toluene

$$P = P_{\text{benzene}}^0 x_{\text{benzene}} + x_{\text{toluene}}^0 P_{\text{toluene}}^0 = 45 + 22.5 = 67.5 \text{ torr.}$$

$$P_{\text{total}} = 67.5 + 10.25 = 77.75 \text{ torr.}$$

2. **Ans. (5.00)**

$$P_{H_2} = \frac{nRT}{V} = \frac{1 \times 0.083 \times 273}{2.27} = \frac{22.7}{2.27} = 10 \text{ bar}$$

$$4 \times 10^{-9} = \frac{p_{H_2} \times [OH^-]}{[I^-]^2} \Rightarrow [OH^-] = 10^{-5}$$

3. **Ans. (4.00)**4. **Ans. (5.00)**5. **Ans. (72.00 to 74.00)**

$$n_{O_3} = \frac{PV}{RT} = \frac{1 \times 2.463}{0.0821 \times 273} = 0.1 \text{ mole}$$

$$1 = K_H \frac{0.1}{0.1 + \frac{360}{18}} \Rightarrow K_H = 201$$

$$6 = K_H \frac{n}{n + \frac{900}{18}} \Rightarrow n = \frac{300}{195} = \frac{20}{13}$$

$$\text{Mass of } O_3 = \frac{20}{13} \times 48 = 73.85 \text{ g}$$

6. **Ans. (5.00)**

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

Sol. $(\theta_1 + \theta_2) \left(\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} \right) = 1$

$$\cos^4 x + \sin^4 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x = 1$$

$$1 - 2\sin^2 x \cos^2 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x = 1$$

$$\left(\sqrt{\frac{\theta_2}{\theta_1}} \cos^2 x - \sqrt{\frac{\theta_1}{\theta_2}} \sin^2 x \right)^2 = 0$$

$$\tan^2 x = \frac{\theta_2}{\theta_1}$$

$$\theta_2 = \theta_1 \tan^2 x$$

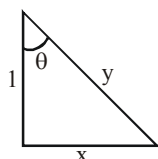
$$\frac{d\theta_2}{d\theta_1} = \tan^2 x$$

2. **Ans. (A,B,C,D)**

Sol. $y \sin \theta = x$

$$y \cos \theta = 1$$

$$(y^n - x^n) = \frac{1 - \sin^n \theta}{\cos^n \theta}$$



$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{\cos \theta} = 0, \quad \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 \theta}{\cos^2 \theta} = 1$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^3 \theta}{\cos^3 \theta} = \infty \Rightarrow \text{Limit does not exist.}$$

3. **Ans. (C,D)**

$$I = \int \frac{(x^3 + 1)(x^3 - 2)}{x^3 \cdot x^2 \sqrt{\left(x + \frac{1}{x^2}\right)^2 + 3}} dx = \int \frac{\left(x + \frac{1}{x^2}\right) \left(1 - \frac{2}{x^3}\right)}{\sqrt{\left(x + \frac{1}{x^2}\right)^2 + 3}} dx$$

$$\text{Put } \left(x + \frac{1}{x^2}\right)^2 + 3 = t^2$$

$$2 \left(x + \frac{1}{x^2}\right) \left(1 - \frac{2}{x^3}\right) dx = 2t dt$$

$$\therefore I = \int \frac{tdt}{t} = t + C = \sqrt{\left(x + \frac{1}{x^2}\right)^2 + 3} + C$$

$$f(x) = \sqrt{\left(x + \frac{1}{x^2}\right)^2 + 3}$$

$$f(1) = \sqrt{7}$$

$$f(-1) = \sqrt{3} \text{ \& } f(2) < f(3)$$

4. **Ans. (B,C)**

$$I_n = \int_0^1 (1 - x^2)^n dx \text{ Put } x = \cos \theta$$

$$\Rightarrow I_n = \int_0^{\pi/2} \sin^{2n+1} \theta d\theta$$

use walli's formula

$$I_n = \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdot \frac{2n-4}{2n-3} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdots (1)$$

$$I_{n+1} = \frac{2n+2}{2n+3} \cdot \frac{2n}{2n+1} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdots (2)$$

$$\frac{I_{n+1}}{I_n} = \frac{2n+2}{2n+3} \text{ and } (1) \Rightarrow I_k = \sum_{r=1}^k \frac{2r}{2r+1}$$

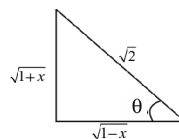
Also in I_n , each fraction is less than 1
so $I_n < 1 \forall n \in \mathbb{N}$

5. **Ans. (A,B)**

$$\begin{aligned} \text{Sol. } \sin^{-1} \sqrt{\frac{1+x}{2}} - \sqrt{2-x} &= \cot^{-1}(\tan \sqrt{2-x}) - \sin^{-1} \sqrt{\frac{1-x}{2}} \\ \Downarrow \quad \quad \quad \Downarrow \quad \quad \quad \Downarrow \quad \quad \quad \Downarrow \\ -1 \leq x \leq 1 \quad x \leq 2 \quad \quad \quad x \leq 2 \quad \quad \quad -1 \leq x \leq 1 \end{aligned}$$

Domain : $x \in [-1, 1]$

$$\sin^{-1} \sqrt{\frac{1+x}{2}} + \sin^{-1} \sqrt{\frac{1-x}{2}} - \cot^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$



$$\frac{\pi}{2} - \cot^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$

$$\tan^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$

$$\text{Let } \sqrt{2-x} = \alpha \text{ where } \alpha \in [1, \sqrt{3}]$$

$$\Rightarrow \tan^{-1}(\tan \alpha) = \alpha \quad \dots\dots(1)$$

$$\text{Case I : } \alpha \in \left[1, \frac{\pi}{2}\right)$$

$$\tan^{-1}(\tan \alpha) = \alpha \text{ happens } \forall \alpha \in \left[1, \frac{\pi}{2}\right)$$

$$1 \leq \sqrt{2-x} < \frac{\pi}{2} \text{ or } 2 - \frac{\pi^2}{4} < x \leq 1$$

$$\text{Case II: } \alpha \in \left[\frac{\pi}{2}, \sqrt{3}\right)$$

$$\tan^{-1}(\tan \alpha) \text{ becomes } \alpha - \pi$$

$$\text{Hence (1) becomes } \alpha - \pi = \alpha \text{ (Not possible)}$$

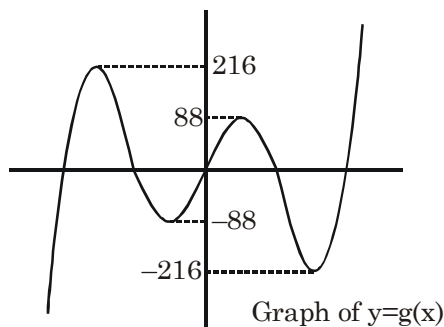
6. **Ans. (A,C,D)**

Sol. For A, D, assume $g(x) = x^n f(x)$

For C; apply LMVT in $[0, 4]$ and $[4, 5]$

7. **Ans. (A,B,C)**

Sol. Let $g(x) = 3x^5 - 50x^3 + 135x$



$$g'(x) = 0 \text{ for } x = \pm 1, \pm 3$$

$$g(1) = 88; g(-1) = -88; g(3) = -216;$$

$$g(-3) = 216$$

$$\text{Now consider } g(x) = a$$

8. **Ans. (A,C)**

Sol. Differentiating $fg = 1$,

$$\text{we get } f'g + g'f = 0 \Rightarrow \frac{f'}{f} = -\frac{g'}{g}$$

Differentiating again,

$$\text{we get } f''g + 2f'g' + g''f = 0$$

$$\Rightarrow \frac{f''}{f'} \cdot \frac{g}{g'} + 2 + \frac{g''}{g'} \cdot \frac{f}{f'} = 0$$

$$\Rightarrow \frac{f''}{f'} \cdot \frac{g}{g'} + 2 - \frac{g''}{g'} \cdot \frac{g}{g'} = 0 \Rightarrow \frac{g}{g'} \left(\frac{f''}{f'} - \frac{g''}{g'} \right) = -2$$

$$\Rightarrow \left(\frac{f''}{f'} - \frac{g''}{g'} \right) = \frac{-2g'}{g} \text{ or } \frac{2f'}{f}$$

Solution for 9 & 10

9. **Ans. (A)**

10. **Ans. (D)**

$$\text{Sol. } f(x) = x^3 - 6x^2 - 36x + 2$$

$$f'(x) = 3x^2 - 12x - 36 = 3(x+2)(x-6)$$

$$f''(x) = 6x - 12 = 6(x-2)$$

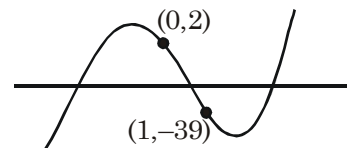
$$\text{(I) } f'(x) > 0 \therefore x \in (-\infty, -2) \cup (6, \infty)$$

$$\text{(II) } x = 6$$

$$\text{(III) } f(x) = 0 \Rightarrow x^3 - 6x^2 - 36x + 2 = 0$$

$$\therefore f(0) = 2 \text{ and } f(1) = -39$$

$\Rightarrow$ three real roots



$$\text{(IV) for rolles theorem } f(a) = f(0)$$

$$a^3 - 6a^2 - 36a + 2 = 2$$

$$\therefore a = 0, \frac{6 + \sqrt{180}}{2}, \frac{6 - \sqrt{180}}{2}$$

$$\therefore a > 0 \Rightarrow k = \frac{6 + \sqrt{180}}{2}$$

Solution for 11 & 12

11. **Ans. (B)**

12. **Ans. (D)**

Sol. (I) $x = 1$

$$14 - 7y + y^2 = 2$$

$$y^2 - 7y + 12 = 0$$

$$y = 4 \text{ or } y = 3$$

$$28x - 7y - 7xy' + 7yy' = 0_{(1,4)}$$

$$M_1 = y' = \frac{7y - 28x}{2y - 7x} \bigg|_{(1,4)} = 0$$

Tangent $y = 4$

$$M_2 = y'_{(1,3)}$$

$$= \frac{-7}{-1} = 7$$

$$y - 3 = 7(x - 1)$$

$$\left(\frac{8}{7}, 4\right)$$

(II) Take log & differentiate

(III) Point of intersection is $x = 1$

now proceed

$$\text{(IV) } \frac{dy}{dx} = 0 \text{ or } \infty$$

SECTION-II

1. Ans. 0.52 or 0.53

$$\begin{aligned}\frac{dy}{dx} &= \left((\sin^{-1} x)^2 - \pi (\cos^{-1} x) \right) \cdot \frac{1}{\sqrt{1-x^2}} \\ &= \left((\sin^{-1} x)^2 - \pi \left(\frac{\pi}{2} - \sin^{-1} x \right) \right) \cdot \frac{1}{\sqrt{1-x^2}} \\ &= \left((\sin^{-1} x)^2 + \pi \sin^{-1} x - \frac{\pi^2}{2} \right) \frac{1}{\sqrt{1-x^2}} \\ &= \left(\left(\sin^{-1} x + \frac{\pi}{2} \right)^2 - \frac{3\pi^2}{4} \right) \frac{1}{\sqrt{1-x^2}}\end{aligned}$$

$$\frac{dy}{dx} = 0 \Rightarrow \sin^{-1} x + \frac{\pi}{2} = \pm \frac{\sqrt{3}}{2} \pi$$

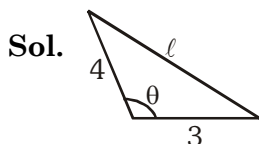
$$\sin^{-1} x = \sqrt{3} \frac{\pi}{2} - \frac{\pi}{2} \text{ only}$$

$$\Rightarrow \frac{\pi}{2} - \cos^{-1} \alpha = \sqrt{3} \frac{\pi}{2} - \frac{\pi}{2}$$

$$\Rightarrow \pi - \cos^{-1} \alpha = \sqrt{3} \frac{\pi}{2}$$

$$\Rightarrow \frac{\cos^{-1}(-\alpha)}{\pi} = \frac{\sqrt{3}}{2}$$

2. Ans. 7.00



Let angle between the hands be θ and distance between tips be ℓ

$$\ell = \sqrt{25 - 24 \cos \theta}$$

we have to find ℓ when $\frac{d\ell}{d\theta}$ is maximum

$$\frac{d\ell}{d\theta} = \frac{12 \sin \theta}{\sqrt{25 - 24 \cos \theta}} \text{ to maximum } \frac{d\ell}{d\theta}$$

$$\frac{d^2\ell}{d\theta^2} = 12 \left\{ \frac{\sqrt{25 - 24 \cos \theta} \cdot \cos \theta - \frac{24 \sin^2 \theta}{2\sqrt{25 - 24 \cos \theta}}}{(25 - 24 \cos \theta)} \right\} = 0$$

$$(25 - 24 \cos \theta) \cos \theta - 12 \sin^2 \theta = 0$$

$$12 \cos^2 \theta - 25 \cos \theta + 12 = 0$$

$$\cos \theta = \frac{3}{4} \quad \cos \theta = \frac{4}{3} \text{ (Not possible)}$$

$$\text{hence } \ell = \sqrt{25 - 24 \cdot \frac{3}{4}} = \sqrt{7}$$

3. Ans. 1.00

Sol. $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ ($f(0) = 0$)

$$f'(x) = 1 + x + x^2 + x^3 + x^4 \quad (f'(0) = 1)$$

$$f''(x) = 1 + 2x + 3x^2 + 4x^3 \quad (f''(0) = 1)$$

$$\text{Now } f(g(x)) = x$$

$$\text{Diff. w.r.to } x$$

$$f'(g(x))g'(x) = 1 \quad \dots(i)$$

$$\text{again Diff. w.r.to } x$$

$$f''(g(x))(g'(x))^2 + g''(x)f'(g(x)) = 0$$

$$x = 0$$

$$f''(g(0))(g'(0))^2 + g''(0)f'(g(0)) = 0 \quad \dots(ii)$$

$$\text{as } f(0) = 0 \Rightarrow g(0) = 0$$

$$\text{and from (i) } g'(0) = \frac{1}{f'(0)}$$

$$\text{now from (ii)}$$

$$f''(0) \left(\frac{1}{f'(0)} \right)^2 + g''(0)f'(0) = 0$$

$$1 \cdot 1^2 + g''(0) \cdot 1 = 0$$

$$g''(0) = -1$$

$$|g''(0)| = 1$$

4. Ans. 8.00

Sol. Let image of focus from $S(-1, -1)$ on

$3x - y = 8$ is (α, β) which lie on directrix

$$\therefore \frac{\alpha+1}{3} = \frac{\beta+1}{-1} = 2 \quad \therefore (\alpha, \beta) = (5, -3)$$

$(7, 13)$ and $(5, -3)$ is a line which is parallel to axis of parabola whose slope is 8

$$\Rightarrow \text{slope of directrix is } = \frac{-1}{8}$$

$$\text{equation of directrix is } y + 3 = -\frac{1}{8}(x - 5)$$

$$8y + x + 19 = 0$$

$$\Rightarrow \frac{b}{a} = 8.00$$

5. Ans. 2.71 or 2.72

Sol. $a_1 = 1, a_n = n(1 + a_{n-1})$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{a_1}\right) \left(1 + \frac{1}{a_2}\right) \dots \left(1 + \frac{1}{a_n}\right)$$

$$\lim_{n \rightarrow \infty} \frac{(1 + a_1)(1 + a_2) \dots (1 + a_n)}{a_1 a_2 a_3 \dots a_n}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{\left(\frac{a_2}{2}\right) \left(\frac{a_3}{3}\right) \dots \left(\frac{a_{n+1}}{n+1}\right)}{a_1 a_2 a_3 \dots a_n} = \frac{a_{n+1}}{(n+1)!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{(n+1)(1 + a_n)}{(n+1)!} = \frac{1}{n!} + \frac{a_n}{n!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{n(1 + a_{n-1})}{n!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{1}{(n-1)!} + \frac{a_{n-1}}{(n-1)!}$$

Continuing in same way;

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \dots + \frac{1}{1!} + \frac{a_1}{1!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} \dots + \frac{1}{n!} \quad (\text{In}$$

reverse order)

As $n \rightarrow \infty$, series on RHS converges to e

6. Ans. 2.00

$$I_1 = \int_0^{\pi} f(\cos x) dx, I_2 = \int_0^{\pi/2} f(\sin x) dx$$

put $x = u$ in I_1 , and $\sin x = u$ in I_2

$$\therefore I_1 = \int_{-1}^1 \frac{f(u)}{\sqrt{1-u^2}} du, I_2 = \int_0^1 \frac{f(u)}{\sqrt{1-u^2}}$$

$$I_1 = 2 \int_0^1 \frac{f(u)}{\sqrt{1-u^2}} du \Rightarrow I_1 = 2I_2$$

$$\therefore \frac{I_1}{I_2} = 2$$